

Technical Incident Report: DNS Conflict, SSL Failure, and Subdomain Resolution

Overview

This report documents a real-world incident involving DNS conflicts, SSL certificate failures, and hosting configuration issues arising from a hybrid setup using **Wix** and **Kartra**.

The root cause was a DNS misalignment between the platforms: the root domain was controlled by Wix, while the primary website was hosted on Kartra. This caused SSL issuance failures and “Connection Not Private” errors in browsers.

A new blog subdomain was created to separate services, stabilize DNS, and ensure proper SSL coverage.

Background

In this scenario:

- **Wix** managed the DNS for the root domain (`exampledomain.com`).
- **Kartra** hosted the primary website (`www.exampledomain.com`).

This setup is common for small businesses but can cause conflicts, particularly on lower-tier plans:

Wix Limitations

- Full DNS control is restricted unless a paid plan is active.
- Only one CNAME per domain is allowed, limiting multi-platform setups.
- SSL cannot be issued for a root domain unless properly assigned.

Kartra Limitations

- Requires the **www** CNAME to host the website.
- Cannot issue SSL for domains that are not correctly pointed.
- Misaligned DNS records lead to untrusted certificates.

Resulting configuration issues:

- Root domain (**exampledomain.com**) → pointed to Wix
 - Main site (**www.exampledomain.com**) → pointed to Kartra via CNAME
 - SSL could not be issued for either domain simultaneously
 - Browser warnings appeared: *“Connection Not Private”*
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Symptoms Observed

- Expired SSL certificate on the root domain
 - SSL handshake failures when connecting to both root and **www** subdomains
 - Inconsistent DNS resolution globally (nslookup, DNS Checker)
 - Intermittent “Connection Not Private” errors for users
 - Blog was not yet configured
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Diagnostic Tools & Methods

Operating Systems: Windows 10, Kali Linux

Tools Used:

- `openssl s_client` for SSL handshakes
- `nslookup` for DNS resolution verification
- DNS Checker for global propagation
- SSL Labs Server Test
- SSL Checker
- WHOIS lookup for nameserver validation

Platforms Involved:

- Wix (DNS host)
 - Kartra (website builder & hosting)
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Findings

DNS Analysis

- Root domain (`exampledomain.com`) pointed to Wix → could not issue SSL
- `www.exampledomain.com` pointed to Kartra → SSL could not issue for root domain
- Wix only allows one CNAME per domain → conflict blocked SSL issuance

SSL Diagnostics

- OpenSSL confirmed expired certificates on the root domain

- SSL Labs scan showed untrusted server certificate for root domain
 - SSL Labs scan for [www](#) subdomain on Kartra showed partial success but inconsistent propagation
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Corrective Actions

Blog Subdomain Creation

- Blog hosted as [blog.exampledomain.com](#) on Wix
- Required paid Wix plan to access DNS editor
- CNAME added for blog subdomain
- DNS and SSL verified via DNS Checker and SSL Checker
- Result: blog subdomain fully operational, SSL issued, independent of root/main site

Attempts to Fix Root Domain Without Transfer

- Tested redirects and page-level forwarding in Kartra
 - All failed due to Kartra plan limitations (requires root domain assignment)
 - Conclusion: root domain needed transfer to Kartra for SSL to function
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Permanent Solution

1. Transfer root domain ([exampledomain.com](#)) to Kartra → allows SSL issuance and renewal
2. Keep blog subdomain ([blog.exampledomain.com](#)) hosted on Wix → isolated DNS and SSL

3. Downgrade Wix plan after transfer → reduces cost, no functional impact
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Lessons Learned / Platform Limitations

- Wix low-tier plans restrict DNS management and SSL control
 - Kartra requires root domain or [www](#) subdomain control to issue SSL
 - Hybrid setups require careful planning to avoid conflicts
 - Subdomains can be used to separate services and bypass platform limitations
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Limitations

Due to client privacy and security considerations, screenshots of DNS records, SSL test results, and platform dashboards have not been included in this report.

Including these screenshots would have required displaying domain-specific information, IP addresses, and platform account details, which could compromise client confidentiality.

All technical findings, remediation steps, and results are fully documented in the text of this report, providing a complete overview of the incident and resolution. While screenshots would have provided visual confirmation, omitting them ensures the report remains safe to share publicly, such as in a GitHub portfolio.

Conclusion

The issues affecting `exampledomain.com` were caused by DNS conflicts between Wix and Kartra, outdated records, and failed SSL renewal. By systematically verifying DNS records, checking SSL status, and testing hosting configurations, the root cause was identified and resolved.

Some upgrade-related changes were unavoidable. Upgrading Wix allowed DNS access and subdomain creation, enabling the blog to function independently. This eliminated the SSL conflict between the root/main domain and Kartra.

The final configuration is stable, secure, and follows best practices:

- DNS integrity restored
- SSL functioning correctly
- Blog subdomain provides flexibility without risking future disruption